cleanly fit the space you provide, and you want those images to grow or shrink if the user changes the size of the window.

For all these examples, we'll use the same method described in listing 8.9 to create the `imageArray` dataset that we'll use. The figures in this chapter will have slight variation from the results of running this code, because we randomly generate some of the visual elements. We can create our first gallery with surprisingly little code.

Listing 8.10 Resizing eight-image gallery

```
imgPerLine = 8;
d3.select("canvas").remove();
d3.select("#traditional")
    .append("div").attr("class", "gallery")
    .selectAll("img").data(imageArray).enter().append("img")
    .attr("class", "infinite")
    .attr("src", function(d) {return d.data;});

redrawGallery();

function redrawGallery() {
    var newWidth = parseFloat(d3.select("div.gallery")
        .node().clientWidth);
    var imageSize = newWidth / imgPerLine;
    function imgX(x) {
      return Math.floor(x / imgPerLine) * imageSize;
    };
    function imgY(x) {
      return Math.floor(x%imgPerLine * imageSize);
    };
    d3.selectAll("img")
        .style("width", newWidth / imgPerLine)
        .style("top", function(d) {return imgX(d.x)})
        .style("left", function(d) {return imgY(d.x)})
};

window.onresize = function(event) {
   redrawGallery();
};
```

Resizes automatically to fit any number of images per row

Deletes the canvas element because it's not needed anymore

Placement code in a separate function for ease of use with dynamic updates

Size based on the parent div width

x and y based on custom accessor functions

Resizes the gallery whenever the page is resized

As shown in figure 8.9, this produces a scrollable div with eight images per line. The images not only scale to fit the div, but rescale as you adjust your browser window.

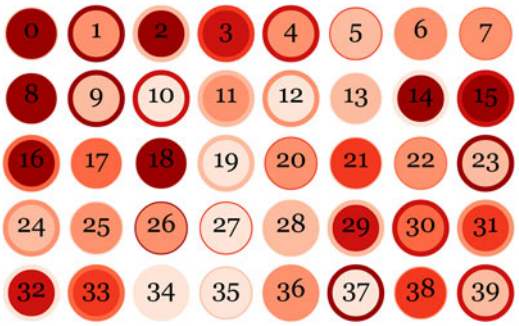


Figure 8.9 Automatically scaled-to-fit images that pack eight images per row